



The natural response of an RC circuit

Capacitor voltage is continuous:

$$V(0^+) = V(0^-) = V_0$$

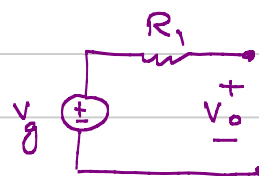
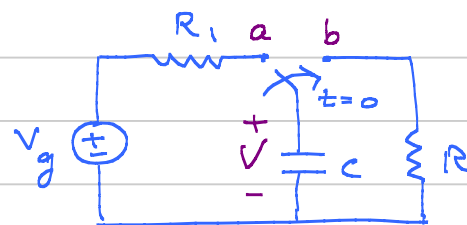
Apply KCL in the ckt. after switching:

$$\frac{V}{R} + C \frac{dV}{dt} = 0$$

$$\frac{dV}{V} = -\frac{1}{RC} dt \rightarrow \int_{V_0}^{V(t)} \frac{dV}{V} = -\frac{1}{RC} \int_0^t dt$$

$$\ln \frac{V(t)}{V_0} = -\frac{t}{RC}$$

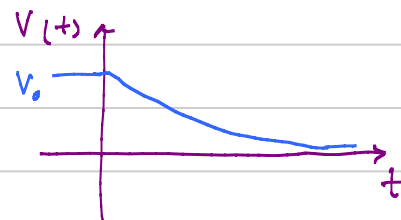
$$V(t) = V_0 e^{-\frac{t}{\tau}}, \quad \tau = RC$$



ckt. at $t = 0^-$
(sw at position 'a', capacitor open)



after switching ($t > 0$)



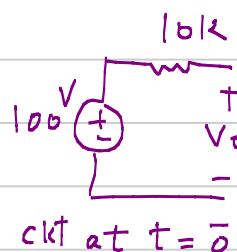
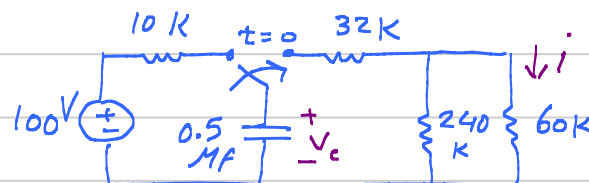
Ex. Find $v_c(t)$ and $i(t)$ for $t \geq 0$.

Use the ckt. before switching

(at $t = 0^-$), with the open ckted capacitor

to find the initial capacitor voltage.

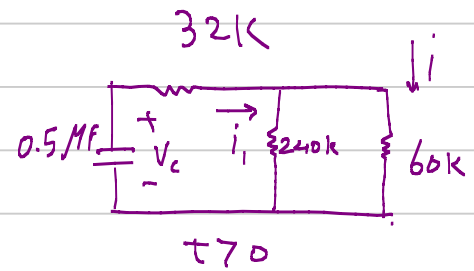
$$V_0 = V(0^-) = 100V$$



ckt at $t = 0^-$

use the ckt. after switching to find τ .

$$R_{eq} = 32k + (240k // 60k) \\ = 80 k\Omega$$



$$\tau = R_{eq} \cdot C = (80 \times 10^3) (0.5 \times 10^{-6}) = \underline{40 \text{ ms}}$$

$$V_c(t) = V_0 e^{-t/\tau} = 100 e^{-\frac{t}{.04}} = \underline{100 e^{-25t} \text{ V}}$$

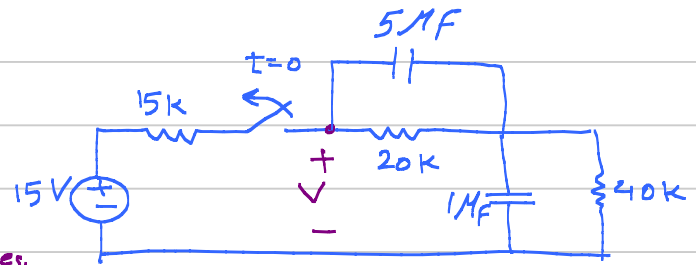
$$i_1 = \frac{V_c}{R_{eq}} = 1.25 e^{-25t} \text{ mA}$$

$$i = i_1 \frac{240}{300} = \underline{e^{-25t} \text{ mA}}$$

Ex. Find $v_c(t)$ for $t \geq 0$.

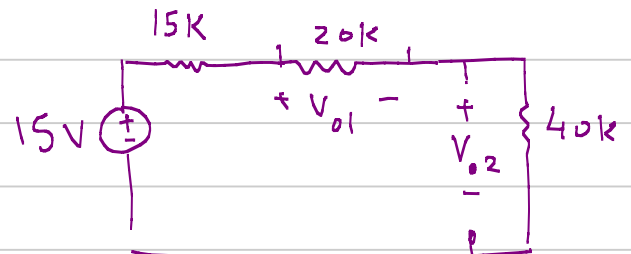
open the capacitors in the ckt.

before switching to find the initial values.



$$V_{01} = 15 \frac{20}{15+20+40} = \underline{4V}$$

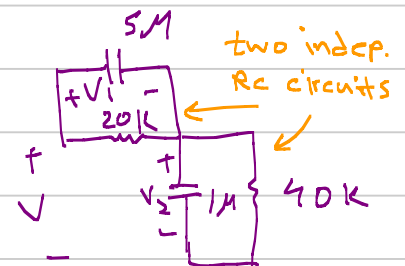
$$V_{02} = 15 \frac{40}{15+20+40} = \underline{8V}$$



$t = 0^-$

$$\tau_1 = (20k)(5M) = 0.1 \text{ s}$$

$$V_{c1}(t) = 4 e^{\frac{-t}{0.1}} = \underline{4 e^{-10t}}$$



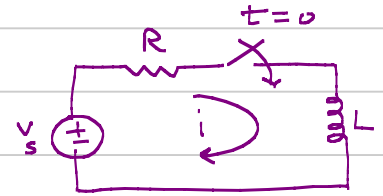
$$\tau_2 = (40k)(1M) = 0.04 \text{ s}$$

$$V_{c2}(t) = 8 e^{\frac{-t}{0.04}} = \underline{8 e^{-25t}}$$

$$V = V_{c1} + V_{c2} = \underline{4 e^{-10t} + 8 e^{-25t}}$$

The step response of an RL circuit

The behavior of the circuit after being connected to a source is called step response.



$$V_s = Ri + L \frac{di}{dt}$$

$$\frac{di}{dt} = \frac{V_s - Ri}{L} = -\frac{R}{L} \left(i - \frac{V_s}{R} \right)$$

$$di = -\frac{R}{L} \left(i - \frac{V_s}{R} \right) dt$$

$$\frac{di}{i - \frac{V_s}{R}} = -\frac{R}{L} dt \rightarrow \int_{I_0}^{i(t)} \frac{di}{i - \frac{V_s}{R}} = -\frac{R}{L} \int_0^t dt$$

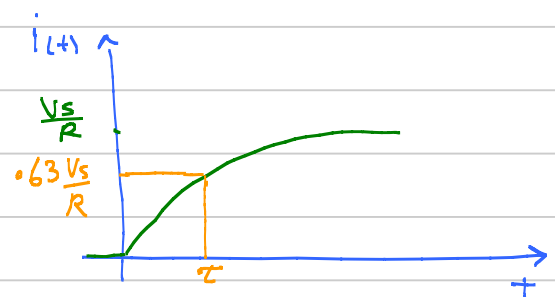
$$\ln \frac{i(t) - \frac{V_s}{R}}{I_0 - \frac{V_s}{R}} = -\frac{R}{L} t \Rightarrow i(t) = \left(I_0 - \frac{V_s}{R} \right) e^{-\frac{R}{L} t} + \frac{V_s}{R}$$

$$i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R} \right) e^{-\frac{t}{\tau}}, \quad \tau = \frac{L}{R}$$

(An arrow points from the term $I_0 - \frac{V_s}{R}$ to the text "Initial Value". Another arrow points from the term $\frac{V_s}{R}$ to the text "Final Value".)

$$\text{If } I_0 = 0 : i(t) = \frac{V_s}{R} (1 - e^{-\frac{t}{\tau}})$$

$$\text{at } t = \tau : i(t) = 0.63 \frac{V_s}{R}$$

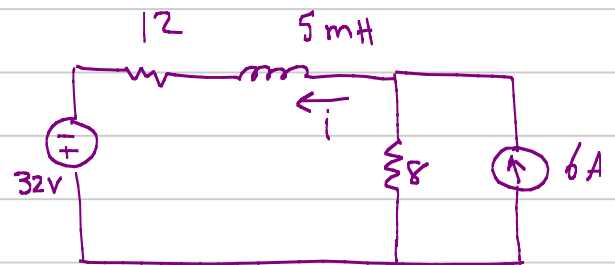
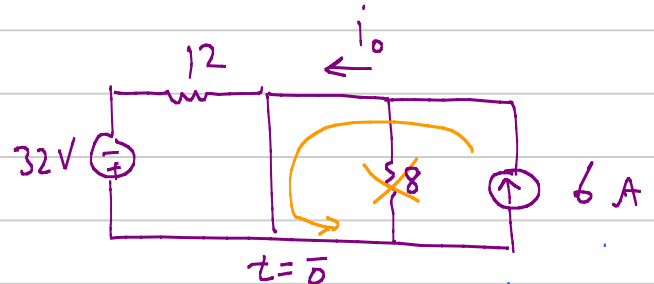
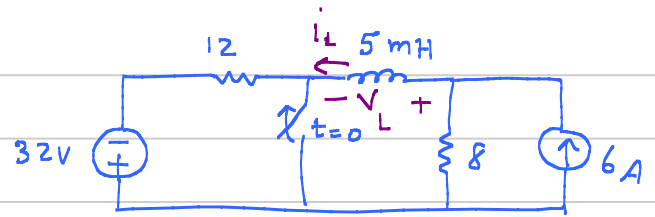


Ex. Find $v_L(t)$ for $t \geq 0$.

Initial value in the ckt. before switching :

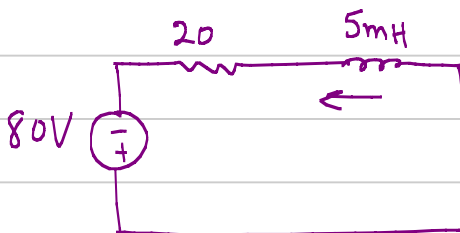
$$i_0 = 6A$$

$$\tau = \frac{L}{R} = \frac{5mH}{20} = 0.25ms$$

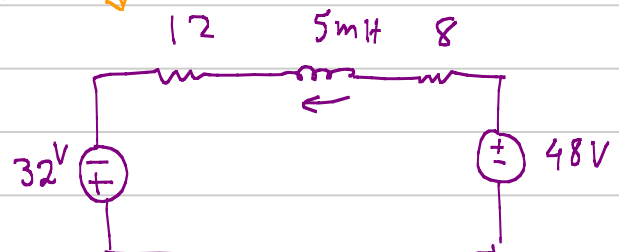


Source transformation

$t > 0$



Combining



At $t = \infty$, inductor is short ckt. again:

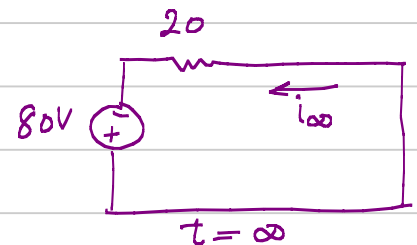
$$\frac{V_s}{R} = \frac{80}{20} = 4A$$

$$i_L = \frac{V_s}{R} + (i_0 - \frac{V_s}{R}) e^{-\frac{t}{\tau}}$$

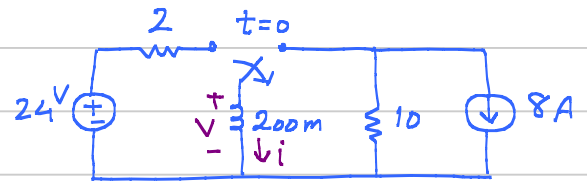
$$= 4 + (6 - 4) e^{-\frac{t}{0.25m}}$$

$$= 4 + 2 e^{-4000t} A$$

$$V(t) = L \frac{di}{dt} = (5m) (-8000 e^{-4000t}) = -40 e^{-4000t} V$$

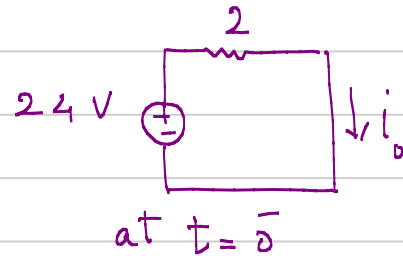


Ex. Find a) $i(0^+)$, b) $v(0^+)$, c) time constant τ , d) $i(t)$, and e) $v(t)$.



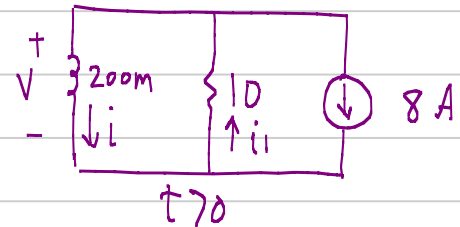
$$i(0^-) = \frac{24}{2} = 12 \text{ A}$$

$$i(0^+) = i(0^-) = \underline{12 \text{ A}}$$



$$i_1(t=0^+) = i(0^+) + 8 = 12 + 8 = 20 \text{ A}$$

$$V_{(0^+)} = V_{10\Omega} = -10 i_1 = \underline{-200 \text{ V}}$$

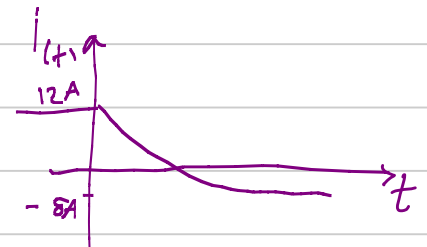
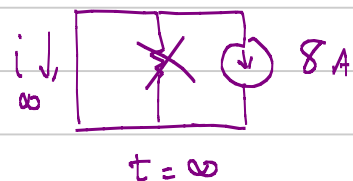


$$c) \tau = \frac{L}{R_{eq}} = \frac{200\text{m}}{10} = \underline{20 \text{ ms}}$$

$$d) i_{(\infty)} = -8 \text{ A}$$

$$i(t) = -8 + (12 - (-8)) e^{\frac{-t}{20\text{ms}}}$$

$$i(t) = -8 + 20 e^{-50t} \text{ A}$$



$$e) V_{(t)} = -10(i(t) + 8) = 10(-8 + 20e^{-50t} + 8) = -200e^{-50t} \text{ V}$$

